Important Notice

This file contains the rubrics and solution of Assignment 2.

The given solution and rubric are based on one valid approach to solve the problem. However, there can be multiple correct methods or variations to reach the same result. Marks will be awarded for any solution that is logically correct and mathematically valid, even if the approach differs from the one presented here.

Question 1.1.1. (6 marks)

Suppose there are finitely many prime numbers $p_1, p_2, \ldots, p_n$ for some natural number $n \geq 2$. Prove or disprove the following:
For every $I \subseteq \{1, 2, \ldots, n\}$ and $J \subseteq \{1, 2, \ldots, n\}$ satisfying $I \neq \varnothing, J \neq \varnothing, I \cap J = \varnothing$ and $I \cup J = \{1, 2, \ldots, n\}$,
the natural number

$$\prod_{i \in I} p_i + \prod_{j \in J} p_j$$

is prime.
Here, $\prod_{i \in I} p_i$ and $\prod_{j \in J} p_j$ denote the product of prime numbers whose indices are in I and J, respectively.
PROOF:
Assume, for contradiction, that the expression $\prod_{i \in I} p_i + \prod_{j \in J} p_j$ is not prime.

$$\exists p_q \in \{p_1, p_2, \ldots, p_n\} \text{ such that } p_q \mid \left(\prod_{i \in I} p_i + \prod_{j \in J} p_j \right) \quad \text{(by Fact 2.1)}.$$

$$\Rightarrow \prod_{i \in I} p_i + \prod_{j \in J} p_j = p_q k \quad \text{for some integer } k.$$

CASE 1: $q \in I$

$$\prod_{j \in J} p_j = p_q k - \prod_{i \in I} p_i.$$

Let I' denote the set obtained from I by removing p_q. Then we can rewrite this as

$$\prod_{j \in J} p_j = p_q \left(k - \prod_{i' \in I'} p_{i'} \right).$$

Hence, p_q divides $\prod_{j \in J} p_j$, that is,

$$p_q \mid \prod_{j \in J} p_j.$$

Using Fact 2.2, this implies $p_q \mid p_m$ for some $m \in J$. Since each prime has only two divisors (1 and itself), we must have $p_q = p_m \Rightarrow q = m \Rightarrow q \in J$. But this contradicts the fact that I and J are disjoint, i.e., $I \cap J = \phi$.
CASE 2: $q \in J$ By a symmetric argument to that in Case 1, we would arrive at $q \in I$, which again violates the assumption that $I \cap J = \phi$. Since both cases lead to contradictions, our assumption must be false. Therefore, the expression

$$\prod_{i \in I} p_i + \prod_{j \in J} p_j$$

must indeed be prime.
NOTE: No other case is possible except $q \in I$ or $q \in J$, because the primes are drawn from the finite set $\{p_1, p_2, \ldots, p_n\} = I \cup J$.

Rubric

The question carries a total of 6 marks, distributed as follows:

- 1 mark for correctly writing the statement for contradiction.
- 1.5 marks for correctly applying **Fact 2.1**.
- 1.5 marks for correctly using **Fact 2.2** to deduce that $p_q = p_m$.
- 2 marks for correct and complete analysis of both cases ($q \in I$ and $q \in J$).

Question 1.1.2. (4 marks). Let A be a finite set and R be an equivalence relation on A. Prove that

$$|A| = \sum_{[a]_R \in A/R} \big|[a]_R\big|,$$

where $\sum_{[a]_R \in A/R} \big|[a]_R\big|$ means the sum of cardinalities of all equivalence classes of A with respect to R.

Proof. Let A be a finite set and R an equivalence relation on A. Since R is an equivalence relation, for every element $a \in A$, the equivalence class $[a]_R = \{\, x \in A \mid (x, a) \in R \,\}$ is non-empty because $a \in [a]_R$ by reflexivity. Thus, every element of A belongs to at least one equivalence class.
Moreover, for any two elements $a, b \in A$, their equivalence classes are either completely equal or completely disjoint. Indeed, if $[a]_R \cap [b]_R \neq \emptyset$, then there exists $x \in A$ such that $x \in [a]_R$ and $x \in [b]_R$. This implies $(x, a) \in R$ and $(x, b) \in R$. By symmetry and transitivity of R, it follows that $(a, b) \in R$, hence $[a]_R = [b]_R$. Otherwise, if no such x exists, then $[a]_R \cap [b]_R = \emptyset$. Therefore, the collection of equivalence classes $A/R = \{[a]_R : a \in A\}$ forms a partition of A; that is, the classes are mutually disjoint and their union equals A:

$$A = \bigcup_{[a]_R \in A/R} [a]_R, \quad \text{where } [a]_R \cap [b]_R = \emptyset \text{ whenever } [a]_R \neq [b]_R.$$

Since A is finite and can be written as a disjoint union of its equivalence classes, the cardinality of A equals the sum of the cardinalities of these classes:

$$|A| = \sum_{[a]_R \in A/R} |[a]_R|.$$

□

Rubric

The question carries a total of 4 marks, distributed as follows:

- 1 mark for showing that any subset in A/R is non-empty.
- 1 mark for showing that every element of A belongs to some equivalence class in A/R.
- 1 mark for proving that any two equivalence classes are either completely equal or have no elements in common.
- 1 mark for concluding that A/R forms a partition of A, and hence the given claim is true.

Question 1.2

1. Solution:

Let R be the set of all sets that are not members of themselves (sometimes called the *Russell set*).

Then there exist two possible cases:

Case 1: R is a set that does not contain itself.

By the definition of R, if R is a set that does not contain itself, then it must belong to the set of all sets that are not members of themselves, i.e., $R \in R$. But this implies that R contains itself, which contradicts the assumption of this case.

Case 2: R is a set that contains itself.

Then, by the definition of R, since R contains itself, it cannot belong to the set of all sets that are not members of themselves, i.e., $R \notin R$. This again leads to a contradiction.

Thus, both assumptions lead to contradictions, and such a set R cannot exist. This is known as Russell's Paradox.

Rubric:

Intro: 1
Case 1: 2
Case 2: 2

2. Solution:

Claim. For every integer $n \geq 2$ there exists a propositional formula u in variables $p_1, \ldots, p_n$ such that exactly

$$\left|\{(a_1, \ldots, a_n) \in \{T, F\}^n : u(a_1, \ldots, a_n) = T\}\right| = 2^{n-1} + 1.$$

Construction and proof. Consider the formula

$$u = p_1 \lor (p_2 \land p_3 \land \cdots \land p_n).$$

Count how many assignments make u true.

- Every assignment with $p_1 = T$ makes u true. There are 2^{n-1} such assignments (the other $n-1$ variables are free).
- Of the assignments with $p_1 = F$, the only one that makes u true is the assignment where $p_2 = \cdots = p_n = T$, because then the conjunct $p_2 \land \cdots \land p_n$ is true. That is exactly one more assignment.

Thus the total number of satisfying assignments is $2^{n-1} + 1$, as required. (The formula is valid for every $n \geq 2$; for $n = 1$ the expression $p_2 \land \cdots \land p_n$ is not present, which is why the problem excludes $n = 1$.)

An alternative general construction is to take any set $S \subseteq \{T, F\}^n$ of size $2^{n-1} + 1$ and write u as the disjunction (DNF) of the characteristic conjunction for each assignment in S:

$$u = \bigvee_{(a_1,\ldots,a_n)\in S} \bigwedge_{i=1}^{n} \begin{cases} p_i & \text{if } a_i = T, \\ \neg p_i & \text{if } a_i = F. \end{cases}$$

This yields a formula true on exactly the assignments in S.

Rubric:

- **(1 mark)** *Claim / statement of existence.* Student clearly states the goal (that such a formula exists for arbitrary $n > 1$).
- **(2 marks)** *Correct formula or constructive method.* Student gives a correct explicit formula (e.g. $p_1 \lor (p_2 \land \cdots \land p_n)$) **or** a correct general constructive method (DNF listing of the desired assignments).
  - Full 2 marks for the simple compact formula above, or a correct DNF construction.
  - 1 mark if the student gives an attempt at construction but it is incomplete or slightly incorrect.
- **(2 marks)** *Correct counting / justification.* Student correctly counts the satisfying assignments of the given formula and concludes the count is $2^{n-1} + 1$.
  - Full 2 marks for the two-step counting (all assignments with $p_1 = T$ = 2^{n-1}; exactly one with $p_1 = F$ makes the conjunction true).
  - 1 mark if the student's counting argument is right in spirit but lacks clarity or a small detail.

Solution for Question 1.3

Question 1.3: Prove or Disprove

1. Power Set Union (2.5 marks)

Statement: Let A, B be sets. Then, $\text{pow}(A \cup B) = \text{pow}(A) \cup \text{pow}(B)$.

Answer: Disprove (False).

Proof/Disproof: The statement is false. We provide a counterexample. Let $A = \{a\}$ and $B = \{b\}$.

1. Calculate $\text{pow}(A \cup B)$:
$$A \cup B = \{a, b\}$$
$$\text{pow}(A \cup B) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$$
2. Calculate $\text{pow}(A) \cup \text{pow}(B)$:
$$\text{pow}(A) = \{\emptyset, \{a\}\}$$
$$\text{pow}(B) = \{\emptyset, \{b\}\}$$
$$\text{pow}(A) \cup \text{pow}(B) = \{\emptyset, \{a\}, \{b\}\}$$

Since $\{a, b\} \in \text{pow}(A \cup B)$ but $\{a, b\} \notin \text{pow}(A) \cup \text{pow}(B)$, the sets are not equal:
$$\text{pow}(A \cup B) \neq \text{pow}(A) \cup \text{pow}(B)$$

2. Equivalence Relations: Intersection and Union (5 marks)

Statement: Let R, S be two equivalence relations on a non-empty set A. Then, $R \cap S$ and $R \cup S$ are also equivalence relations on A.

Answer: Partially True ($R \cap S$ is true, $R \cup S$ is false).

Part A: $R \cap S$ is an equivalence relation (True)

We show $R \cap S$ is reflexive, symmetric, and transitive.

1. **Reflexive:** Since R and S are reflexive, for all $a \in A$, $(a, a) \in R$ and $(a, a) \in S$. Thus, $(\mathbf{a}, \mathbf{a}) \in \mathbf{R} \cap \mathbf{S}$.
2. **Symmetric:** Assume $(a, b) \in R \cap S$. Then $(a, b) \in R$ and $(a, b) \in S$. Since R and S are symmetric, $(b, a) \in R$ and $(b, a) \in S$. Thus, $(\mathbf{b}, \mathbf{a}) \in \mathbf{R} \cap \mathbf{S}$.
3. **Transitive:** Assume $(a, b) \in R \cap S$ and $(b, c) \in R \cap S$.
   - Since $(a, b) \in R$ and $(b, c) \in R$, by transitivity of R, $(a, c) \in R$.
   - Since $(a, b) \in S$ and $(b, c) \in S$, by transitivity of S, $(a, c) \in S$.

   Thus, $(\mathbf{a}, \mathbf{c}) \in \mathbf{R} \cap \mathbf{S}$.

Part B: $R \cup S$ is an equivalence relation (False)

We provide a counterexample showing $R \cup S$ may fail transitivity. Let $A = \{1, 2, 3\}$.

- Let $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$
- Let $S = \{(1, 1), (2, 2), (3, 3), (2, 3), (3, 2)\}$

Then $R \cup S$ contains $(1, 2)$ and $(2, 3)$. For $R \cup S$ to be transitive, $(1, 3)$ must be in $R \cup S$. However, $(1, 3) \notin R$ and $(1, 3) \notin S$, so $(\mathbf{1}, \mathbf{3}) \notin \mathbf{R} \cup \mathbf{S}$. Thus, $R \cup S$ is not transitive, and not an equivalence relation.

3. Relation Defined by a Function (2.5 marks)

Statement: Let R be a relation on A defined by $(a, b) \in R$ if and only if $F(a) = F(b)$ for a function $F : A \to B$. Then, R is an equivalence relation.

Answer: Prove (True).

Proof: We show R is reflexive, symmetric, and transitive.

1. **Reflexive:** For any $a \in A$, $F(a) = F(a)$ is true. By definition of R, $(\mathbf{a}, \mathbf{a}) \in \mathbf{R}$.
2. **Symmetric:** Assume $(a, b) \in R$. By definition, $F(a) = F(b)$. Since equality is symmetric, $F(b) = F(a)$. By definition of R, $(\mathbf{b}, \mathbf{a}) \in \mathbf{R}$.
3. **Transitive:** Assume $(a, b) \in R$ and $(b, c) \in R$. By definition:

$$F(a) = F(b) \quad \text{and} \quad F(b) = F(c)$$

By the transitivity of equality, $F(a) = F(c)$. By definition of R, $(\mathbf{a}, \mathbf{c}) \in \mathbf{R}$.

Since R satisfies all three properties, R is an equivalence relation.

Rubric for Question 1.3 (Total: 10 Marks)

Q1.3.1: Power Set Union (2.5 Marks)

0.5 Marks: States the verdict: **FALSE** (Disprove).

1.0 Mark: Provides a correct **counterexample** (e.g., $A = \{a\}, B = \{b\}$).

1.0 Mark: Shows the **inequality** by identifying an element (e.g., $\{a, b\}$) present in $\text{pow}(A \cup B)$ but missing in $\text{pow}(A) \cup \text{pow}(B)$.

Q1.3.2: Equivalence Relations: Intersection and Union (5.0 Marks)

Part A: $R \cap S$ is an Equivalence Relation (2.5 Marks)

0.5 Marks: States the verdict: **TRUE**.

0.5 Marks: Correctly shows **Reflexivity**.

0.5 Marks: Correctly shows **Symmetry**.

1.0 Mark: Correctly shows **Transitivity**.

Part B: $R \cup S$ is NOT an Equivalence Relation (2.5 Marks)

0.5 Marks: States the verdict: **FALSE**.

1.0 Mark: Provides a valid **counterexample** (e.g., relations on $A = \{1, 2, 3\}$).

1.0 Mark: Clearly shows the **failure of Transitivity** in the union for the chosen counterexample.

Q1.3.3: Relation Defined by a Function (2.5 Marks)

0.5 Marks: States the verdict: **TRUE** (Prove).

2.0 Marks:
- **0.5** Shows **Reflexivity** ($F(a) = F(a)$).
- **0.5** Shows **Symmetry** ($F(a) = F(b) \implies F(b) = F(a)$).
- **1.0** Shows **Transitivity** (using transitivity of equality).

Question 1.4 (10 marks). Let A be a non-empty set and R be an arbitrary equivalence relation on A. Let A/R be the set of all equivalence classes of A with respect to R, and G : A → A/R be the function such that for every a ∈ A, G(a) := [a]R. Let B be a non-empty set and F : A → B be an arbitrary function that satisfies the property that if (a, b) ∈ R then F(a) = F(b). Prove that there exists a unique function H : A/R → B such that F = H ◦ G. Here, = refers to the equality of functions.

Rubric Distribution (Total: 10 Marks)

1. Existence of Function H (6 Marks)

• Clearly defines **H([a]) = F(a)**. — *1 mark*

H:A/R→B by H([a]R) = F(a) for all a∈A.

• Demonstrates that **H is properly constructed** from F using equivalence classes. — *2 marks*

If two elements a and b belong to the same equivalence class, then they are related by R (i.e., (a,b)∈R.
Since the question says that whenever (a,b)∈R, we have F(a)=F(b) this shows that all elements in the same class give the same F-value.
Hence, the value of H([a]R)=F(a) doesn't depend on which representative a we pick — it's the same for the whole class.
*That's why H is **well-defined**.*

• Proves that **F = H ◦ G**, i.e., for all a ∈ A, **F(a) = H(G(a))**. — *3 marks*

To prove F=H◦G :
For any a∈A
(H◦G)(a)=H(G(a))=H([a]R).
Thus, F=H◦G

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2. Uniqueness of Function H (4 Marks)

• Assumes two functions **H_1, H_2 : A/R → B** satisfy **F = H_1 ◦ G** and **F = H_2 ◦ G**. *2marks*
• Shows that for any equivalence class [a], **H_1([a]) = H_2([a])**. — *2 marks*

Assume two functions H1,H2:A/R→B satisfy F=H1◦G=H2◦G.
Let [a]R∈A/R.
Then H1([a]R)=(H1◦G)(a)=F(a)=(H2◦G)(a)=H2([a]R)
Thus H1([a]R)=H2([a]R) for all [a]R∈A/R so H1=H2
*Hence, H is **unique**.*

This is one of the correct approaches to solve the problem. Other logically valid and mathematically sound solutions will also be accepted and given full credit.

Problem 1.5 — Solutions and Marking Rubric

Total: 10 marks

(1) 3 marks (2) 7 marks

Question 1

(3 marks) Let A, B be non-empty sets and $F : A \to B$ be a function. For every $y \in B$, define

$$F^{-1}(y) := \{x \in A : F(x) = y\}.$$

Let $\mathcal{P} := \{F^{-1}(y) : y \in B\}$. Does $\mathcal{P}$ form a partition of A? Prove your answer.

Solution

Recall: A collection $\mathcal{C}$ of subsets of A is a *partition* of A iff

1. every element of $\mathcal{C}$ is nonempty,
2. elements of $\mathcal{C}$ are pairwise disjoint,
3. the union of all sets in $\mathcal{C}$ equals A.

We check these properties for $\mathcal{P}$.

(Union equals A). Take any $x \in A$. Then $F(x) \in B$, hence $x \in F^{-1}(F(x))$. Therefore

$$\bigcup_{y \in B} F^{-1}(y) = A.$$

(So property (3) holds.)

(Pairwise disjoint). Let $y_1, y_2 \in B$ with $y_1 \neq y_2$. If $x \in F^{-1}(y_1) \cap F^{-1}(y_2)$ then $F(x) = y_1$ and $F(x) = y_2$, contradicting $y_1 \neq y_2$. Thus

$$F^{-1}(y_1) \cap F^{-1}(y_2) = \varnothing.$$

(So property (2) holds.)

(Nonempty blocks). If $y \in B$ is not in the image $\mathrm{Im}(F)$, then $F^{-1}(y) = \varnothing$. So some members of $\mathcal{P}$ may be empty unless F is surjective.

Conclusion. The collection $\mathcal{P}$ satisfies union-equals-A and pairwise disjointness always, but it contains only nonempty sets exactly when F is surjective. Hence

$$\boxed{\mathcal{P} \text{ is a partition of } A \iff F \text{ is surjective.}}$$

Question 2

(7 marks) Is $\mathbb{N} \times \mathbb{N} \times \mathbb{N}$ countably infinite? Prove your answer.

Solution

We show $\mathbb{N}^3$ is countably infinite.

Method 1: Product of countable sets. It is standard that $\mathbb{N}$ is countably infinite and that $\mathbb{N} \times \mathbb{N}$ is countably infinite (for example by Cantor's pairing function or the diagonal argument). A product of two countable sets is countable; therefore $(\mathbb{N} \times \mathbb{N}) \times \mathbb{N} \cong \mathbb{N}^3$ is countable. Since $\mathbb{N}^3$ is infinite, it is countably infinite.

Method 2: Explicit enumeration by sum (constructive). For each integer $k \geq 0$, define the finite set

$$S_k := \{(a, b, c) \in \mathbb{N}^3 : a + b + c = k\}.$$

Each S_k is finite (there are only finitely many nonnegative integer solutions to $a+b+c = k$). Now list elements of $\mathbb{N}^3$ by increasing k:

$$S_0, S_1, S_2, \ldots$$

Since every triple $(a, b, c) \in \mathbb{N}^3$ belongs to exactly one S_k (with $k = a + b + c$), this yields a countable sequence enumerating all elements of $\mathbb{N}^3$. Thus $\mathbb{N}^3$ is countably infinite.

Conclusion. $\boxed{\mathbb{N} \times \mathbb{N} \times \mathbb{N} \text{ is countably infinite.}}$

Marking Rubric (detailed)

Question 1 (3 marks total)

Criterion	Marks
State definition of partition (nonempty, pairwise disjoint, union = A)	0.5
Show $\bigcup_{y \in B} F^{-1}(y) = A$ (argument: $x \in F^{-1}(F(x))$)	0.75
Show pairwise disjointness (argument by contradiction)	0.75
Discuss nonemptiness and relate to surjectivity; conclude "iff F surjective"	1.0
Total	**3.0**

Question 2 (7 marks total)

Criterion	Marks
State what "countably infinite" means (existence of bijection/enumeration)	0.5
Show $\mathbb{N} \times \mathbb{N}$ is countable (mention diagonal or pairing) or cite known fact	1.0
Argue product with $\mathbb{N}$ remains countable (conclude $\mathbb{N}^3$ is countable)	2.0
Provide an explicit constructive enumeration (e.g., partition by sum $a + b + c = k$ and list finite S_k)	2.0
State conclusion $\mathbb{N}^3$ is countably **infinite** and justify infinitude	1.0
Clarity, notation, and presentation (neatness)	0.5
Total	**7.0**

Solutions and Rubrics

Discrete Mathematics — Quiz / Assignment

Question 1 (4 marks)

Task: Read about partial-order relations and partially ordered sets online and describe them in detail in your own words.

Solution (written in own words)

A *partial order* on a set P is a binary relation $\preceq$ on P that expresses a notion of "comparison" between some elements but not necessarily every pair. Concretely, $\preceq$ must satisfy three properties for all $x, y, z \in P$:

- **Reflexive:** $x \preceq x$. Every element is related to itself.
- **Antisymmetric:** If $x \preceq y$ and $y \preceq x$, then $x = y$. This prevents distinct elements from being mutually comparable in both directions.
- **Transitive:** If $x \preceq y$ and $y \preceq z$, then $x \preceq z$. Comparisons compose.

A set P equipped with a partial order $\preceq$ is called a *partially ordered set*, or *poset*, denoted $(P, \preceq)$.

Intuition and remarks:

- "Partial" means not every pair of elements must be comparable. If every pair is comparable (for all x, y either $x \preceq y$ or $y \preceq x$), the order is called a *total* or *linear* order.
- Examples:
  - $(\mathbb{Z}, \leq)$ is a total order (hence a poset).
  - The power set $(\mathcal{P}(S), \subseteq)$ of any set S with inclusion is a poset; many pairs of distinct subsets are incomparable (neither subset of the other).
  - Divisibility on the positive integers: $a \preceq b$ iff $a \mid b$ is a partial order (not total).
- **Hasse diagrams:** A common way to draw a finite poset is a Hasse diagram: elements are points, and we draw edges only for "covering" relations (direct comparisons with no intermediate element). Edges are drawn so that larger elements appear higher up; reflexive and transitive edges are omitted for clarity.
- **Special concepts:**

- A *chain* is a subset of a poset in which every pair is comparable (a totally ordered subset).
- An *antichain* is a subset where no two distinct elements are comparable.
- A *minimal* element is one that has no strictly smaller element in the set; a *least* element (if it exists) is smaller than every element.
- *Upper bounds*, *least upper bound* (join), *greatest lower bound* (meet) lead to lattice theory when they exist for all pairs.

Short illustrative example: Let $S = \{a, b, c\}$ and define $a \preceq a, b \preceq b, c \preceq c$, plus $a \preceq b$. Then a and b are comparable, but c may be incomparable with a and b; this is a simple poset that is not a total order.

Rubrics for Question 1 (4 marks):

Criteria	**Marks**
Clear definition of partial order (lists reflexive, antisymmetric, transitive) in own words	2.0
Explanation of poset and the meaning of "partial" vs "total"	1.0
At least two correct and distinct examples with brief explanation (e.g., $(\mathcal{P}(S), \subseteq)$, divisibility, $(\mathbb{Z}, \leq)$)	0.5
Mention of related concepts (Hasse diagram, chains/antichains, minimal/least elements or joins/meets)	0.5
Total	**4.0**

Question 2

Given: A, B non-empty sets.

(a) (3 marks)

Statement: Let $F : A \to B$ be a function. Prove or disprove:

$$F \text{ is injective} \iff \forall X \subseteq A,\ \forall Y \subseteq A,\ F(X \cap Y) = F(X) \cap F(Y),$$

where $F(Z) := \{F(z) : z \in Z\}$.

Solution

("$\Rightarrow$") If F is injective, then equality holds for all X, Y. We always have $F(X \cap Y) \subseteq F(X) \cap F(Y)$ because any $a \in X \cap Y$ gives $F(a) \in F(X)$ and $F(a) \in F(Y)$. So the nontrivial direction is $F(X) \cap F(Y) \subseteq F(X \cap Y)$.

Take $b \in F(X) \cap F(Y)$. Then there exist $x \in X$ and $y \in Y$ with $F(x) = b = F(y)$. Since F is injective, $x = y$. Thus this common element x belongs to $X \cap Y$, and $b = F(x) \in F(X \cap Y)$. Hence $F(X) \cap F(Y) \subseteq F(X \cap Y)$. Combining both inclusions gives equality.

("$\Leftarrow$") If equality holds for every $X, Y \subseteq A$, then F is injective. To prove injectivity, let $u, v \in A$ with $F(u) = F(v)$. Consider the singleton sets $X = \{u\}$ and $Y = \{v\}$. Then

$$F(X) = \{F(u)\}, \qquad F(Y) = \{F(v)\} = \{F(u)\},$$

so $F(X) \cap F(Y) = \{F(u)\}$ is nonempty. By hypothesis,

$$F(X \cap Y) = F(\{u\} \cap \{v\}) = F(\varnothing) \text{ if } u \neq v,$$

but $F(\varnothing) = \varnothing$. For equality to hold we must have $X \cap Y \neq \varnothing$, so $\{u\} \cap \{v\} \neq \varnothing$, hence $u = v$. Therefore F is injective.

Conclusion: The equivalence is true.

Rubrics for 2(a) (3 marks):

Criteria	**Marks**
Prove $F(X \cap Y) \subseteq F(X) \cap F(Y)$ (clear statement)	0.5
Injective $\Rightarrow$ equality: correct argument using injectivity to identify preimages	1.0
Equality for all $X, Y \Rightarrow$ injective: correct singleton argument	1.0
Clarity and correctness of logical flow (conclusion)	0.5
Total	**3.0**

(b) (3 marks)

Statement: Let $F : A \to A$ be a function satisfying $F(F(x)) = x$ for every $x \in A$. Prove that F is bijective.

Solution

We show first injectivity, then surjectivity.

Injective: Suppose $F(a) = F(b)$ for some $a, b \in A$. Apply F to both sides:

$$F(F(a)) = F(F(b)) \implies a = b,$$

because $F(F(x)) = x$ for every x. Hence F is injective.

Surjective: Let $y \in A$ be arbitrary. Take $x = F(y)$ (which belongs to A). Then

$$F(x) = F(F(y)) = y.$$

So y has a preimage x. Since y was arbitrary, F is surjective.

Conclusion: F is both injective and surjective, so F is bijective.

Rubrics for 2(b) (3 marks):

Criteria	Marks
Injectivity proof by applying F (clear and correct)	1.5
Surjectivity proof by constructing preimage $x = F(y)$ (clear and correct)	1.0
Conclusion that bijectivity follows	0.5
Total	**3.0**

Overall marks distribution

Question 1 (partial orders)	4.0
Question 2(a) (image of intersection and injectivity)	3.0
Question 2(b) (involution implies bijective)	3.0
Total	**10.0**

Notes for graders:

- For Question 1, reward clarity and correct use of terminology; partial credit for partial definitions or fewer examples.

- For the proofs, require clear logical steps; small algebraic slips may lose minor marks but logical structure should be credited.
- In 2(a) the forward inclusion $F(X \cap Y) \subseteq F(X) \cap F(Y)$ is elementary and should be awarded even if the student forgets to explicitly write it, provided they use its fact in the correct direction.